IN-CLASS QUIZ

Graphs and MSTs
Representations & Kruskal

5 multiple-choice questions · 1.5 minutes each

Vertices are labelled a, b, c, …; numbers on edges are weights.

BEFORE WE START

Scan to answer

Open the form on your phone and keep it open for all five questions.

https://forms.gle/41j5HdsP3QYVbveC8

QUESTION 1

≈ 1.5 minutes

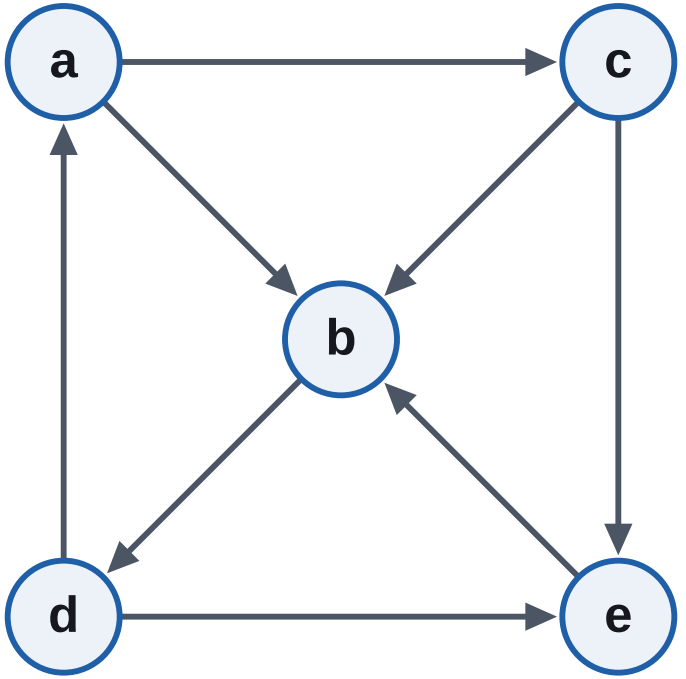
Which is the adjacency-list representation of the directed graph shown?

A a: [d] b: [a, c, e] c: [a] d: [b] e: [c, d]

B a: [b, c] b: [d] c: [b, e] d: [a, e] e: [b]

C a: [b, c, d] b: [a, c, d, e] c: [a, b, e]
d: [a, b, e] e: [b, c, d]

D a: [b, c] b: [d] c: [b, e] d: [a] e: [b, d]



QUESTION 2

≈ 1.5 minutes

An undirected graph with no self-loops has n = 1000 vertices and m = 3000 edges. How many entries of its adjacency matrix are 1, and how many nodes are there in total across all its adjacency lists?

A 3,000 ones; 6,000 list nodes

B 3,000 ones; 3,000 list nodes

C 6,000 ones; 3,000 list nodes

D 6,000 ones; 6,000 list nodes

QUESTION 3

≈ 1.5 minutes

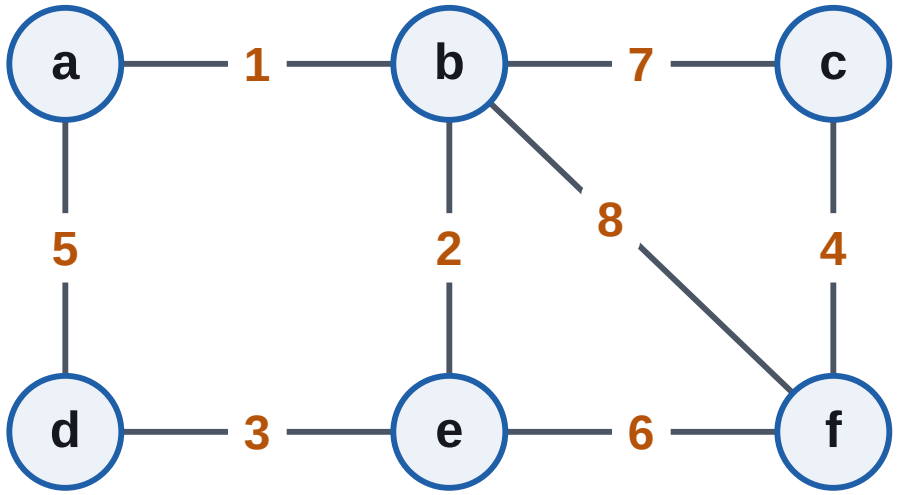
Kruskal's algorithm is run on the weighted graph shown. Which edge does it reject first, and what is the total weight of the MST?

A No edge is rejected before the MST is complete; MST weight 15

B Edge a–d is rejected first; MST weight 16

C Edge a–d is rejected first; MST weight 17

D Edge c–f is rejected first; MST weight 19



QUESTION 4

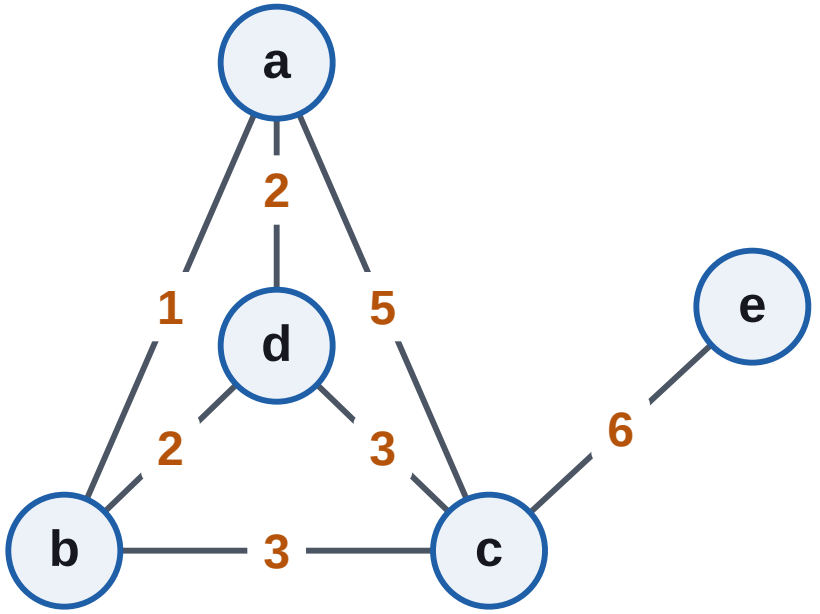
≈ 1.5 minutes

Kruskal's algorithm is run on a connected graph with 10 vertices and 20 edges. At some moment it has accepted 6 edges. At that moment, how many components does the forest built on the accepted edges have (an isolated vertex counts as a tree)? How many more edges will it accept?

- **A** Cannot be determined from the information given; depends on the graph and edges chosen
- **B** 4 components; 4 more edges
- **C** 4 components; 3 more edges
- **D** 1 component; 3 more edges

QUESTION 5

≈ 1.5 minutes

For the weighted graph G shown, which option is entirely correct about the number of distinct MSTs, an edge in every MST, and an edge in no MST?

- **A** 4 MSTs; c–e is in every MST; a–c is in no MST
- **B** 1 MST; a–b is in every MST; a–c is in no MST
- **C** 4 MSTs; a–b is in every MST; c–e is in no MST
- **D** 2 MSTs; c–e is in every MST; a–c is in no MST